

EXPERIMENT 7

Users, Roles and Permissions in Zoho Creator

1. Aim

To examine user management, role creation, and permission configuration in Zoho Creator, and to understand how access privileges are controlled for different users in an application.

2. Introduction

Zoho Creator offers a user access management system through which administrators can decide who can enter an application and which operations they are allowed to perform. It uses Users, Roles, and Permissions to provide role-based access control (RBAC). Users are the people given application access, while roles represent their responsibilities and position in the hierarchy. Permissions specify the operations that users or roles can perform on application modules and records. This organized method supports secure and controlled access in applications used by multiple people.

3. Objectives of the Study

- To gain knowledge of users, roles, and permissions available in Zoho Creator.
- To understand how users are added to an application and assigned suitable roles.
- To study role hierarchy and its effect on access to application data.
- To learn the creation and administration of customized permissions.
- To examine data-sharing options between peers and subordinate users.
- To implement an organized access-control structure for a business application.

4. Procedure

- Sign in to the Zoho Creator platform.
- Open the required application or build a new one.
- Go to Settings and open the User Permissions section.
- Add application users by entering their name and email address.
- Select and assign the suitable role for each user.
- Use the Roles tab to create roles according to the required organizational hierarchy.
- Enter role information including Role Name, Reports To (parent role), peer data sharing, and description.
- Create customized permission profiles through the Permissions tab.
- Configure module-level permissions for each permission profile (Access, View, Edit, Delete).

- Configure the required API and security permission options.
- Link the required permissions with the appropriate users and roles.
- Set Data Sharing options to control which records can be viewed.
- Save the setup and test it using different user accounts to confirm the configuration.
- Publish the application once the user and permission configuration is complete.

5. Illustrative Example

Consider a Food Delivery System where users such as CEO, Admin, Staff, and Customer use the application. Since their responsibilities differ, each user is provided with an appropriate level of access to the data and features.

5.1 Implementation using Zoho Creator

Build a Food Delivery System application and set up the required users, roles, and permissions as given below:

Roles Hierarchy:

- CEO (top-level role)
 - Admin (reports to CEO)
 - Staff (reports to Admin)
 - Customer (reports to Staff)

Permissions Profiles:

- Admin permissions – Admin permissions Profile
- Read – Read permission for all components
- Staff Permissions – Staff Permissions Profile
- Write – Write permission for all components

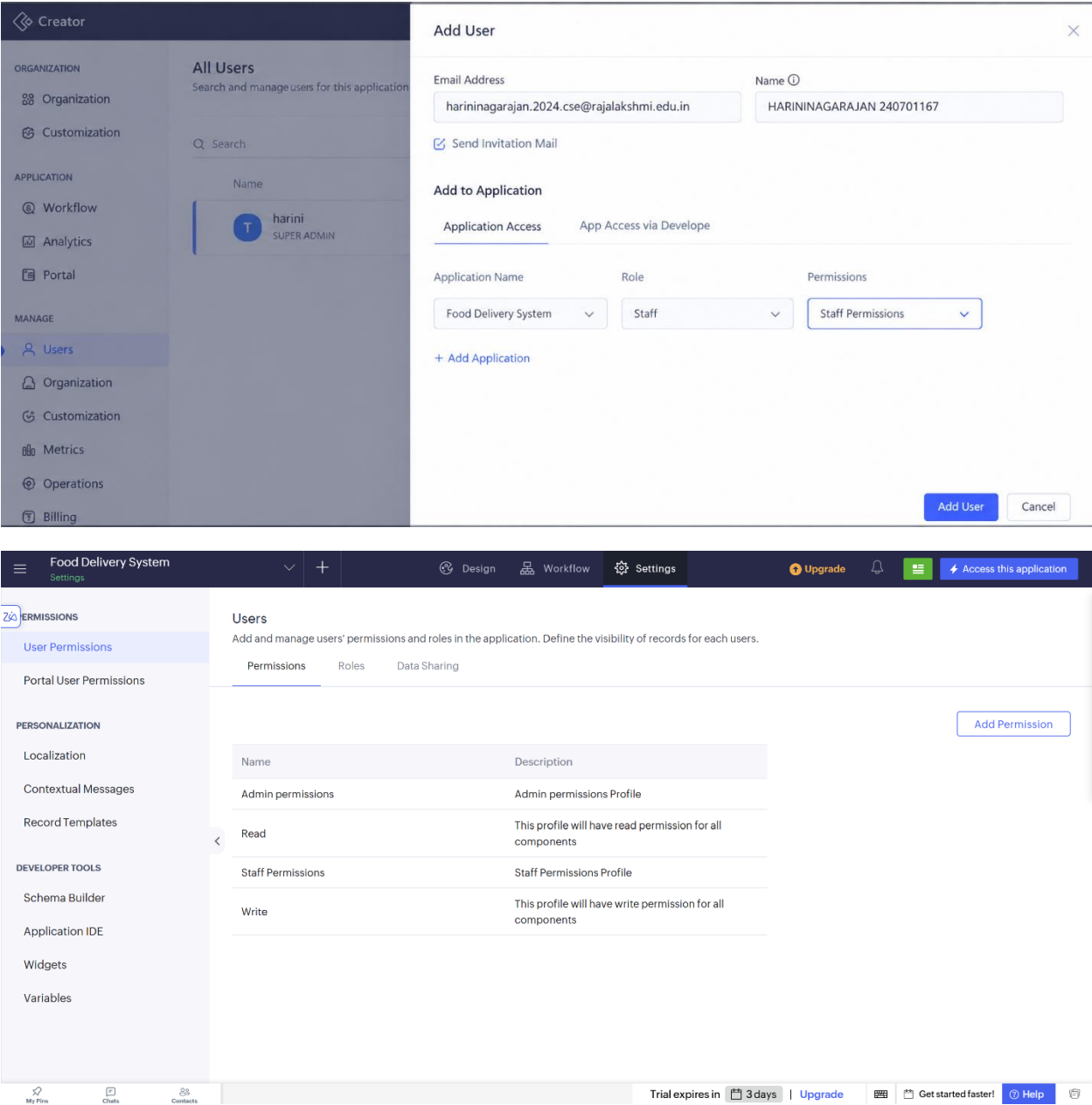
User Assignment:

- User: harininagarajan.2024.cse@rajalakshmi.edu.in
- Name: HARININAGARAJAN 240701167
- Role: Staff
- Permission: Staff Permissions

Module Permissions for Staff Permissions Profile:

- Customer Profile: Access ✓, View (2 Reports), Edit (2 Reports)
- Menu Item: Access ✓, View (2 Reports), Edit (2 Reports)
- Order Line Item: Access ✓, View (All Order Line Items), Edit (All Order Line Items)

5.2 Illustrative Design



Food Delivery System

Settings

DesignWorkflowSettings

Upgrade

Access this application

PERMISSIONS

User Permissions

Portal User Permissions

PERSONALIZATION

Localization

Contextual Messages

Record Templates

DEVELOPER TOOLS

Schema Builder

Application IDE

Widgets

Variables

Users

Add and manage users' permissions and roles in the application. Define the visibility of records for each users.

PermissionsRolesData Sharing

Expand AllCollapse All

Manage Role HierarchyAdd Role

CEO

Admin

Staff

Customer

My Pins

Chats

Contacts

Trial expires in 3 days | Upgrade

Get started faster!

Help

Food Delivery System

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PermissionsRolesData Sharing

Role Details

Role Name: Staff

Reports to: Admin

Share Data with Peers: No

Description: Report to admin

EditDeleteGo To Roles List

Associated Users

Name	Email address
harininagarajan.2024.cse@rajalakshmi.edu.in	harininagarajan.2024.cse@rajalakshmi.edu.in

App Page

Form

Connections

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DesignWorkflowSettings

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Access this application

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Users

Add and manage users' permissions and roles in the application. Define the visibility of records for each users.

PermissionsRolesData Sharing

Name: Staff Permissions

Description: Staff Permissions Profile

API & Security Permission

Expand

Module Permissions

Define the permission for records associated with the forms

Module	Access	View	Edit	Delete	More
Customer Profile	☒	2 Reports	2 Reports	☐	☐
Menu Item	☒	2 Reports	2 Reports	☐	☐
Order Line Item	☒	All Order Line Items	All Order Line Items	All Order Line Items	☐

UpdateCancel

My Pins

Chats

Contacts

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Get started faster!

Help

6. Advantages and Limitations

6.1 Advantages

- Offers an organized role-based method for controlling application access.
- Improves data protection by preventing unauthorized access.
- A role hierarchy makes user responsibilities easier to organize.
- Access permissions can be set for individual users as well as roles.
- Data-sharing options support controlled collaboration among users.
- Grouping users into roles reduces the effort required for administration.

6.2 Limitations

- Complicated permission setups need careful planning and configuration.
- Incorrect permission settings may cause unwanted access or data exposure.
- Larger organizations may require extra customization to meet their needs.
- Changes to the role hierarchy may require users to be assigned again.
- More advanced permission requirements may need Deluge scripting.

7. Result

The experiment was successfully carried out by examining user management, role creation, and permission settings in Zoho Creator. The exercise showed how users, roles, and permissions can work together to provide a secure and organized access-control system for a business application while reducing manual administration.

